

PRUTHVI SHAH

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SUMMARY

Recent Information Technology graduate with hands-on experience building AI and machine learning applications using Python, LLMs, RAG, LangChain, FastAPI, and SQL. Experienced in developing end-to-end ML pipelines, retrieval systems, predictive models, and backend APIs, with a strong interest in Applied AI, Generative AI, and Machine Learning Engineering.

EDUCATION

G H Patel College of Engineering and Technology

B.Tech, Information Technology

May 2026

CGPA: 8.60/10

EXPERIENCE

ML Intern

Jan 2026 – Apr 2026

Thick Code

Rajkot

- Analyzed structured and unstructured datasets across 8 applied ML projects (regression, classification, time-series) under structured biweekly deliverables to identify trends and deliver business insights, performing data preparation, cleansing, feature engineering, and exploratory data analysis (EDA) in Python, NumPy, pandas, and scikit-learn.
- Built and validated predictive models using statistical analysis techniques including ADF stationarity tests and correlation analysis, checking data quality and documenting performance benchmarks to support model selection decisions.

PROJECTS

Football IQ AI | *Python, FastAPI, LangChain, ChromaDB, Sentence Transformers, Groq API, Next.js*

- Built a full-stack Retrieval-Augmented Generation (RAG) application answering natural-language queries grounded in a 31-document knowledge base, achieving 93% exact top-1 retrieval accuracy using sentence-transformer embeddings with top-k similarity search.
- Engineered a structured, source-cited prompt with an explicit low-relevance fallback to reduce LLM hallucination, using Llama 3.3 70B via Groq API as a single grounded inference boundary.
- Shipped an 8-endpoint REST API supporting SSE streaming and history-aware multi-turn conversation, paired with a Next.js chat frontend.

Custom Retrieval Engine | *Node.js, HNSW, KD-Tree, PostgreSQL, pgvector, Vercel Serverless*

- Implemented three vector search algorithms from scratch – HNSW, KD-Tree, and brute-force – including distance metrics, heap-based k-NN, and graph construction, with a live endpoint benchmarking all three on identical queries.
- Designed a grounded RAG pipeline with sliding-window chunking (250 words, 30-word overlap) and a cosine-similarity threshold filter, returning an explicit “not found” response instead of guessing when no chunk qualifies.
- Built an agentic ingestion pipeline that autonomously researches a topic via the Wikipedia API, then chunks, embeds, and stores the resulting knowledge, with PostgreSQL + pgvector persistence across serverless cold starts.

Stock Market Forecasting Dashboard | *Flask, TensorFlow, Keras, statsmodels, scikit-learn, SQLite*

- Led model development on a 4-person team benchmarking ARIMA, Prophet, and LSTM on 1,256 daily observations across multiple tickers; LSTM achieved RMSE 2.89 and MAE 2.23, outperforming ARIMA by 12% and Prophet by 30%.
- Built a preprocessing pipeline with ADF stationarity testing and seasonal decomposition, weighing LSTM’s 12x longer training time (145.6s vs. 12.3s for ARIMA) against accuracy gains to guide model selection tradeoffs.
- Implemented SQLite-backed multi-user authentication and data persistence for the Flask web application, deployed to production on Render.

SKILLS

- Languages:** Python, SQL, JavaScript
- ML & Data Analysis:** Machine Learning, AI/ML, Deep Learning, Neural Networks, LSTM, Predictive Modeling, Predictive Analytics, Classification, Statistical Analysis, Statistics, Data Science, Data Preparation, Data Cleansing, Feature Engineering, Exploratory Data Analysis (EDA), Pandas, NumPy, Scikit-learn, TensorFlow, Keras, PyTorch, statsmodels
- Visualization:** Data Visualization, Matplotlib, Seaborn, Plotly, Streamlit
- GenAI & NLP:** Generative AI, LLMs, RAG, LangChain, LangGraph, Hugging Face, Hugging Face Hub, Vector Databases, Sentence Transformers, ChromaDB, NLP
- Backend & Web:** Flask, FastAPI, React, Next.js, Node.js, REST APIs, MongoDB, SQLite, PostgreSQL
- Cloud & Tools:** AWS, Git, GitHub, Postman, Jupyter

ACHIEVEMENTS

- Top 25/500 Finalist**, CVMU Hackathon – built Medify, an AI medical assistant app
- Secretary**, ISTE GCET Student Branch – organized technical events and managed member operations
- Winner**, Best Business Idea Contest, CVM University
- Winner**, AI Bootcamp, JunoHub

EXTRACURRICULARS

- AWS Academy Graduate** – Cloud Developing (AWS Academy / Coursera)
- Data Science Bootcamp** (2022) – Attended and worked on multiple projects, staying current with industry trends in AI/ML
- Branch Representative, ISA Club** (2023–2025) – Organized events and contributed to community outreach initiatives
- Member, Rotaract Club** (2022 – Jun 2026) – Coordinated events and supported local communities